

PAPER_879: Buoyancy Klein-Gordon Scalar Field EOM

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Abstract

Euler-Lagrange derivation for the buoyancy scalar field $\varphi_{\text{buoy}}(x, t)$ via Klein-Gordon equation with UQFF buoyancy source. The Lagrangian $\mathcal{L} = (1/2)(\partial_{\mu}\varphi)^2 - (1/2)m_{\text{eff}}^2\varphi^2 + J_{\text{buoy}}\cdot\varphi$ yields the EOM $\square\varphi + m_{\text{eff}}^2\varphi = J_{\text{buoy}}$ where m_{eff}^2 encodes the buoyancy mass scale from $\beta_i \cdot \Sigma U_g \cdot \Omega_g \cdot M / (d_g \cdot c^2 \cdot \hbar^2) \cdot [\text{UA}]$.

1. Core Equations

$$\begin{aligned}\square\varphi + m_{\text{eff}}^2\varphi &= J_{\text{buoy}} \\ m_{\text{eff}}^2 &= \beta_i \cdot \Sigma U_g \cdot \Omega_g \cdot M / (d_g \cdot c^2 \cdot \hbar^2) \cdot [\text{UA}] \\ \varphi_{\text{static}}(r) &= (J_{\text{buoy}}/m_{\text{eff}}^2) \cdot [1 - \exp(-m_{\text{eff}} \cdot r)]\end{aligned}$$

2. Parameters

Parameter	Default	Description
M	1.989e30 kg	Central mass
d_g	2.55e20 m	Galactic distance scale
omega_g	7.3e-16 rad/s	Galactic angular frequency
Ug_sum	1e-8 m/s ²	Sum of Ug contributions
F_UBi_over_FU	0.8	Buoyancy-to-field ratio
r	1e16 m	Radial distance

3. UQFF Integration

This calculator integrates `buoyancy_lagrangian_eom.py` from Session 204 into the CP4 pipeline as class #463.

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